

CANBUS BMS Protocol

This document is intended for manufacturers of Managed Batteries: batteries with a BMS inside the communicates with a Victron system. This document describes the used protocol.

1. General

1.1 Connections

The BMS of the battery is connected to the VE.Can port on the [GX-device](#). The GX-device is the central master in a Victron system.

Do not forget the terminator. A bag of two pieces is included in the GX device packaging. They look like this: <https://www.victronenergy.com/accessories/ve-can-rj45-terminator>.

For examples of detailed installer documentation, refer to the [BYD + Victron](#) or the [Pylontech + Victron](#) manual.

A changelog of the Victron firmware in the GX product range is available at <https://professional.victronenergy.com>. Create a login there, go to Firmware → Venus OS, and then see the word document.

1.2 Discharge current not limited in island operation

During a grid failure, or when installed in an off-grid system with a diesel generator, or even without one, the Multis and Quattros will no not limit their max discharge current. The discharge current is in such case defined by the size of the AC loads.

For off-grid systems the installer will (have to) design the system such that inverter capacity including peak loads is within the maximum discharge current of the battery.

1.3 Canbus specs

- Standard frame format, 11-bit identifier, also known as CAN 2.0A
- Both 250kbps and 500kbps are supported. The installer needs to select the correct profile while setting up the system. See Settings → Services → Canbus profile. And select either “VE.Can & CAN-bus BMS (250 kbit/s)”, or “CAN-bus BMS (500 kbit/s)”.

1.4 Communication timeouts

Timeout mechanism implemented in the BMS

In case the BMS / Battery has a safety feature that disconnects the pack with an internal contactor in

case it no longer receives the 0x305 keep alive messages, make sure that the time out is not too short. The minimum advised time out is 10 minutes, needed to cater for firmware updates and reboots of the GX Device.

Timeout mechanism implemented in the Victron system

When the inverter did not receive the 0x351 message from the battery for 3 seconds, the system will stop charging and discharging. Our advice is to make the BMS send this message every second.

1.5 Installer documentation

Once all matching tests have been completed successfully, we'll add the battery and set of notes to our public documentation here: https://www.victronenergy.com/live/battery_compatibility:start

1.6 More information

- GX Device firmware update instructions [here](#).
- Detailed CAN-ID documentation [here](#).

2. Messages sent from BMS to the Inverter

Minimum required CAN-IDs are 0x351, 0x355, 0x356 and 0x35A. All other fields are optional.

CAN ID	Offset (bytes)	Name	Data type	Scaling	Unit
0x351	0	Charge voltage	un16	0.1	V
	2	Max charge current	sn16	0.1	A
	4	Max discharge current	sn16	0.1	A
	6	Discharge voltage	un16	0.1	V
0x355	0	SOC value	un16	1	%
	2	SOH value	un16	1	%
	4	High res SOC (optional)	un16	0.01	%
0x356	0	Battery voltage	un16	0.01	V
	2	Battery current	sn16	0.1	A
	4	Battery temperature	sn16	0.1	C

CAN ID	Offset (bytes)	Name	Data type	Scaling	Unit
0x35A	0 (bit 0+1)	General alarm	(not implemented)		
	0 (bit 2+3)	Battery high voltage alarm			
	0 (bit 4+5)	Battery low voltage alarm			
	0 (bit 6+7)	Battery high temperature alarm			
	1 (bit 0+1)	Battery low temperature alarm			
	1 (bit 2+3)	Battery high temperature charge alarm			
	1 (bit 4+5)	Battery low temperature charge alarm			
	1 (bit 6+7)	Battery high current alarm			
	2 (bit 0+1)	Battery high charge current alarm			
	2 (bit 2+3)	Contactor Alarm	(not implemented)		
	2 (bit 4+5)	Short circuit Alarm	(not implemented)		
	2 (bit 6+7)	BMS internal alarm			
	3 (bit 0+1)	Cell imbalance alarm			
	3 (bit 2+3)	Reserved			
	3 (bit 4+5)	Reserved			
	3 (bit 6+7)	Reserved			
	4 (bit 0+1)	General warning	(not implemented)		
	4 (bit 2+3)	Battery high voltage warning			
	4 (bit 4+5)	Battery low voltage warning			
	4 (bit 6+7)	Battery high temperature warning			
	5 (bit 0+1)	Battery low temperature warning			
	5 (bit 2+3)	Battery high temperature charge warning			
	5 (bit 4+5)	Battery low temperature charge warning			
	5 (bit 6+7)	Battery high current warning			
	6 (bit 0+1)	Battery high charge current warning			
	6 (bit 2+3)	Contactor warning	(not implemented)		
	6 (bit 4+5)	Short circuit warning	(not implemented)		
	6 (bit 6+7)	BMS internal warning			
	7 (bit 0+1)	Cell imbalance warning			
	7 (bit 2+3)	System status (online/offline) [1]			
	7 (rest)	Reserved			
0x35E	0	Manufacturer name	string (8 chars)		
0x35F	0	Battery Model	un16		
	2	Firmware version	un16		
0x370	0	Battery / BMS name part 1	string (8 chars)		
0x371	0	Battery / BMS name part 2	string (8 chars)		
0x372	0	Number of modules OK	un16		
	2	Number of modules blocking charge	un16		
	4	Number of modules blocking discharge	un16		
	6	Number of modules offline	un16		
0x373	0	Min. cell voltage [1]	un16	mV	
	2	Max. cell voltage [1]	un16	mV	
	4	Lowest cell temperature [1]	un16	Kelvin	
	6	Highest cell temperature [1]	un16	Kelvin	
0x374		Min. cell voltage id string [1]	string (8 chars)		

CAN ID	Offset (bytes)	Name	Data type	Scaling	Unit
0x375		Max. cell voltage id string [1]	string (8 chars)		
0x376		Min. cell temperature id string [1]	string (8 chars)		
0x377		Max. cell temperature id string [1]	string (8 chars)		
0x378	0	Energy in [1]	un32	100	Wh
	4	Energy out [1]	un32	100	Wh
0x379	0	Installed capacity [1]	un16	Ah	
0x380	0	Serial number, first 8 characters [1]	string (8 chars)	7-bit ascii	
0x381	0	Serian number, last 8 characters [1]	string (8 chars)	7-bit ascii	

[1]: Victron extension, currently under development.

0x351 - Charge voltage (CVL), discharge current limit (DCL) and charge current limit (CCL) and battery discharge voltage

- Discharge voltage is not used (yet). Instead, the installer is instructed to configure the DC low disconnect voltages according to the battery specification.
- CCL: a dc-coupled ESS system (ie. a system with solar chargers and installed and configured to Victron ESS instructions) cannot handle CCL=0 during operation. So, when the battery is full, don't send CCL=0, but instead send a reduced CVL. Test with the Feedback excess solar setting enabled. A too low CVL will cause the battery to be discharged. A too high limit causes over charging. See also the quirks section here:
<https://github.com/victronenergy/dbus-systemcalc-py/blob/master/delegates/dvcc.py>
- When the grid is disconnected, the maximum discharge current cannot be regulated. Discharge current depends solely on the loads.
- A max discharge current of 0A will disable discharge.
 - If the grid is connected, loads will be powered from the grid. Battery power will not be used.
 - If the grid is disconnected, the Multi will switch off.
 - Charging will still be allowed if max charge current exceeds 0A.
- Both max discharge current and max charge current should be reported as positive values.

0x355 - SOC information

- High res SOC is not supported (yet).

0x356 - Batt voltage, current and temp

- A negative current indicates discharge, positive current indicates charge. Temperature is in degrees centigrade.

0x35A - Warnings and Alarms

In this message, each warning and alarm is implemented to consist of two bits.

Bit 0	Bit 1	
0	0	Alarm or warning cleared (deprecated)
1	0	Alarm or warning raised
0	1	Alarm or warning cleared
1	1	Alarm or warning cleared (deprecated)

Data is shown to the user on the GX Devices, as well as logged to the [VRM Portal](#).

Note that we recommend to use only 10 and 01. Not 00 and 11. In the future, we plan to change the implementation such that it will automatically hide fields reported as 00 or 11 from the user interface. This will help us showing 100% correct information to users and installers for all different batteries and their implementations, without having to hardcode a map of supported fields per battery manufacturer and model, or even their firmware version.

Lastly, note that these fields have no effect on the charging or discharging behaviour of our inverter/chargers, chargers or other products. The only message that is used for control, is 0x351.

Byte 7 bit 2&3 - System status (online/offline)

A simple bit. To indicate if the main relay (or mosfets in certain designs) are enabled or not. Quite often we see while remotely diagnosing a system that it would be nice to see that status. While it can be deduced from alarm flags in some cases, the purpose of this bit is to have the status in an unambiguous way; without having to know the specifics of a certain brand or type of battery.

0x35E - Manufacturer name

- 8 char long name, encoded in 7-bit ascii
- The data will be used to identify the battery, and may be used as battery name (see below for more information)
- If this information is not provided, a generic name will be used. See 2.1 below.

0x35F - Battery type and software versions

- Two byte type ID (not implemented)
- Two byte firmware versions (big-endian ordering, MSB first)

0x370 and 0x371 - Battery / BMS Name (Victron only fields)

- together these two fields can contain a 16 char long name of the battery.
- encoded in 7-bit ascii
- the data is concatenated, starting with 0x370, and appending 0x371 to the right.
- If the name is shorter than 16 characters, the remainder should be padded with zeros.

0x372 - Module status

This field has been designed such that it works both for battery types that have MosFETs in each battery module as well as types that have contactors (or even both).

The intention behind this field is to help installers and support engineers do initial troubleshooting on systems that are not performing OK.

Send 0xFFFF for not supported fields. I.e a battery with only a Contactor will send 0xFFFF for field 2 and 3.

0x373 to 0x377 - Min/max cell voltage and temperature

0x373 contains the data.

- For any unsupported fields, send either 0x0000, 0xFFFF, or, for example when not sending temperature, shorten the message using a lower DLC (Data Length Code).

0x374 to 0x377 is for identification:

- The strings are to be encoded in 7-bit ascii. Unused characters to be set to 0.
- Free format: up to the battery manufacturer to put something in here that helps an engineer to identify which cell or unit is giving problems.
- Use the string to allow a service engineer to identify which package/module/cell is giving a low or high voltage.

0x378 - Total energy in and out

- Two 32-bit values for energy in (charge) and energy out (discharge).
- These counters are used to generate the data on our VRM dashboard. See example below.
- It is recommended that these values are stored in non-volatile storage.
- If storage is not feasible, the inverter algorithm used to generate graphs on VRM can handle resetting counters.
- 100Wh units are used because this gives the best balance between resolution and maximum value.
 - A maximum energy value of 42MWh can be recorded.
 - If this value is exceeded, the counter should roll over to zero.



0x379 - Installed capacity

This is different from 0x35F. In 0x35F in bytes 4 & 5 the usable capacity in Ah is given as a UN16. At 48V this translates to maximum possible capacity of just over 3MWh, which should be sufficient for the foreseeable future. It is recommended that batteries send the usable online capacity: if one or more modules have disconnected, send a lower total usable capacity.

The purpose of 0x379 is to show the installed capacity, so that the service engineer can see how much capacity is offline.

0x380 and 0x381 - Serial number

This field is optional and may not be usable for all battery manufacturers. Some example usages are:

- Systems with a single BMS
- For batteries that have their own cloud-function, send an ID that allows the service technician to identify the same battery on the Battery Manufacturer Cloud
- The definition is the same as for 0x370 and 0x371, and the cell voltage and temperature id fields.
- Free format

2.1 Battery name

There are several sources used by the GX Device to determine the name of the battery. In order of priority:

- The data of CAN-IDs 0x370 and 0x371 if present.
- The data of CAN-ID 0x35E if present and recognised. The data may be used literally, but may also be replaced by a more readable name.
- If the GX Device identifies the battery by some other means, a hard coded name will be used.
- If all else fails, the name will be *CAN-bus BMS battery*.

3. Messages sent by the Inverter to the BMS

0x305 [Keepalive from inverter to BMS]

Sent to the battery every second by the CCGX (data consists of 8 bytes, all zeros).

0x307 [Inverter identification from inverter to BMS]

Sent to the battery every second. This message contains the ASCII string 'VIC' in bytes 4 to 6.

Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
0x12	0x34	0x56	0x78	V	I	C	0x00

Byte 7 is reserved for future use.

4. FAQ

Q1: Do you need to see the 0x359 frame (warnings and alarms) to operate?

No, that is a battery manufacturer specific CAN-ID which we implemented because that manufacturer does not support 0x35A. For all other manufacturers, we prefer 0x35A over 0x359.

5. Verifying implementation

After implementing this protocol in your BMS, follow these steps to verify proper working:

5.1 Connection and basic communication

Connect and configure the CCGX:

1. Connect the battery to the CCGX, on the VE.Can port.
 - Pin 3 is GND
 - Pin 7 is CAN-H
 - Pin 8 is CAN-L
2. In the CCGX Menu, go to Settings → Services → CAN-bus profile, and depending on the CAN baudrate of your BMS select either:
 1. VE.Can & CAN-bus BMS (250 kbit/s)
 2. CAN-bus BMS (500 kbit/s)

3. Next, the battery should show up in the Device list



In case it does not, here are some tips that help identify the problem:

First of all, login to the CCGX with ssh. Instructions [here](#)

Then use `candump` to see the data being communicated on the port. Note than on a Venus-GX you should replace `can0` with `can1`.

This is what typical CAN communications look like:

```
root@ccgx:~# candump can0
interface = can0, family = 29, type = 3, proto = 1
<0x354> [8] 04 c0 00 1f 03 00 00 00
<0x35a> [8] 00 00 00 00 00 00 00 00
<0x351> [8] 45 02 4c 04 4c 04 00 00
<0x355> [8] 59 00 64 00 00 00 00 00
<0x356> [8] 31 16 ce 00 3e 01 00 00
<0x305> [8] 00 00 00 00 00 00 00 00
<0x354> [8] 04 c0 00 1f 03 00 00 00
<0x35a> [8] 00 00 00 00 00 00 00 00
<0x351> [8] 45 02 4c 04 4c 04 00 00
<0x355> [8] 59 00 64 00 00 00 00 00
<0x356> [8] 31 16 d1 00 3e 01 00 00
<0x305> [8] 00 00 00 00 00 00 00 00
<0x354> [8] 04 c0 00 1f 03 00 00 00
<0x35a> [8] 00 00 00 00 00 00 00 00
<0x351> [8] 45 02 4c 04 4c 04 00 00
<0x355> [8] 59 00 64 00 00 00 00 00
<0x356> [8] 31 16 d0 00 3e 01 00 00
<0x305> [8] 00 00 00 00 00 00 00 00
<0x354> [8] 04 c0 00 1f 03 00 00 00
```



```
<0x35a> [8] 00 00 00 00 00 00 00 00
```

The message <0x35a> is the one used by the driver to recognize that there is battery connected.

In case you see 29 bit identifiers, change the code in the BMS to use 11 bit identifiers instead. 29 bit identifiers look like this in candump:

```
<0x00000356> [8] 14 79 01 62 00 f0 02 40
<0x1510a8a0> [8] 00 00 00 00 10 05 07 01
<0x161088b9> [8] 03 01 00 00 00 00 00 04
<0x305> [8] 00 00 00 00 00 00 00 00      <--- sent by the CCGX, that's
why this one is right and the other are not :)
```

You can also look at the output of the driver. But that doesn't say much:

```
root@ccgx:~# cat /log/lg-resu-interface/current | tai64nlocal
2016-06-29 02:06:27.860565500 *** CCGX booted (1) ***
2016-06-30 06:57:03.917694500 *** CCGX booted (1) ***
```

5.2 Derating and charge voltages

Go into the menu, and check that the four dynamic limits are correct. In this example, for example the battery low voltage is not available:



5.3 Alarms

Generate the alarms on the BMS, and double check that they appear on the CCGX.

6. Changes

2019-09-19

- Add new fields

2019-08-19

- Update warnings and alarms documentation
 - complete the list in the large table; including ones known in the industry but not implemented by us.
 - explain that we prefer 01 and 10 for sending warnings and alarms

2019-08-06

- Update the document to reflect the present situation:
 - The Multi will switch off if there is no grid and the BMS requests a zero discharge current

- We support more alarms now.
- The way you enable the CAN-bms driver has changed.

2019-05-06

- Updated contents today's operation (DVCC is available now) & updated introduction with more links.

2017-12-28

- Added warning about soon coming DVCC and SVS improvements.

2017-09-04

- Align canbus specs text with the new canbus profiles options that are in Venus since v2.07.

2017-02-13

- Improved wording in Chapter 1.
- Added 0x35F battery type and firmware version

2017-02-01

- Since CCGX v2.02, VE.Direct Solar chargers will be disabled when the BMS sends max discharge current = 0 or when the BMS stops sending messages.

2016-11-21

- Removed 100% grid parallel restriction, as this was fixed in CCGX v1.72. Note that it still only works icm the Hub-4 Assistant.

2016-10-24 - v1.72

- Fixed bug related to alarms and warnings coming from LG and Sony batteries. Alarms were interpreted as warnings, warnings as alarms.

2016-10-10

- Change battery to inverter keepalive message from 0x35A to 0x351. Will be part of the Venus v1.80 release
- Improve wording in Chapter 1.

2016-09-06

- In CCGX version 1.70 and 1.71, all alarms would disable charge and discharge, except the low battery alarm, which would disable discharge only. In all other versions (older and newer), the alarms will not have a direct effect on charging or discharging. Instead, the CCGX firmware will always follow the maximum charge limit and maximum discharge current limit, even in alarm situations.

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